The Short-Time Remainder Is One Order Higher

Let C be a finite-dimensional matrix with

$$H := \frac{C + C^*}{2} \geq 0, \qquad F(t) := \|e^{-Ct}\|_2^2.$$

If C is hypocoercive with index m, the known leading-order result gives

$$F(t) = 1 - c\, t^{2m+1} + O(t^{2m+2}), \qquad c > 0, \tag{1}$$

as $t \downarrow 0$. The point of this note is that, when $m \geq 1$, the remainder in (1) is automatically one order better: there is no term of degree $2m + 2$. The first order at which a new coefficient can appear is $2m + 3$.

The mechanism is a parity property of the logarithmic norm.

Lemma 1 (Odd logarithmic normal form)**.** *For every matrix C there is a real-analytic function μ, defined near 0, such that*

$$\log F(t) = t\, \mu(t^2) \tag{2}$$

for all sufficiently small $t > 0$. In particular, the right-hand Taylor series of $\log F(t)$ contains only odd powers of t.

Proof. Set

$$Q(t) := e^{-C^* t} e^{-Ct}, \qquad Z(t) := \operatorname{Log} Q(t).$$

Thus $Q(t)$ is positive definite, $Z(t)$ is Hermitian, and $\log F(t) = \lambda_{\max}(Z(t))$. The identity

$$Q(-t) = e^{C^* t} Q(t)^{-1} e^{-C^* t}$$

implies, by analytic functional calculus,

$$Z(-t) = -e^{C^* t} Z(t) e^{-C^* t}. \tag{3}$$

Take the right polar decomposition $e^{C^* t} = U(t)R(t)$, where $U(t)$ is unitary and $R(t) > 0$. Both sides of (3) are Hermitian. Consequently $R(t)Z(t)R(t)^{-1}$ is Hermitian, which gives $R(t)^2 Z(t) = Z(t)R(t)^2$. Since $R(t) > 0$, it follows that $R(t)$ commutes with $Z(t)$, and hence

$$Z(-t) = -U(t)Z(t)U(t)^*. \tag{4}$$

The polar unitary of an inverse is the adjoint, so $U(-t) = U(t)^*$. Near $t = 0$, let $V(t) := U(t)^{-1/2}$ be the analytic principal inverse square root and put $\widehat{Z}(t) := V(t)^* Z(t) V(t)$. Since $V(-t) = U(t)V(t)$, equation (4) yields

$$\widehat{Z}(-t) = -\widehat{Z}(t).$$

Therefore $\widehat{Z}(t) = tB(t^2)$ for a real-analytic Hermitian matrix family B. By analytic Hermitian perturbation theory, the eigenvalues of $B(s)$ can be parametrized by real-analytic branches. After shrinking the interval, one branch $\mu(s)$ is maximal for every small $s > 0$. Since $\widehat{Z}(t)$ is unitarily similar to $Z(t)$,

$$\log F(t) = \lambda_{\max}(Z(t)) = t\, \lambda_{\max}(B(t^2)) = t\, \mu(t^2),$$

which proves (2). $\square$

Proposition 1 (The missing even term)**.** *Assume* (1) *and* $m \geq 1$*. Then, for some real number* b*,*

$$F(t) = 1 - c\,t^{2m+1} + b\,t^{2m+3} + O\Big(t^{2m+5} + t^{4m+2}\Big)\,. \tag{5}$$

In particular, the coefficient of t^{2m+2} *is zero and*

$$F(t) = 1 - c\,t^{2m+1} + O(t^{2m+3}).$$

Thus $2m+3$ *is the next* admissible *order for an independent correction. The coefficient* b *may itself vanish for a special matrix.*

Proof. Since $\log(1+x) = x + O(x^2)$, equation (1) shows that the leading term of $\log F(t)$ is $-c\,t^{2m+1}$. Lemma 1 excludes every even power, so for some $b \in \mathbb{R}$,

$$\log F(t) = -c\,t^{2m+1} + b\,t^{2m+3} + O(t^{2m+5}).$$

Exponentiating gives (5): the first nonlinear contribution is the square of the leading logarithmic term, hence has degree $4m+2$. Because $m \geq 1$, this degree is strictly larger than $2m+2$, and the claimed vanishing follows. □

For the first genuinely hypocoercive index, $m = 1$, the statement is especially transparent:

$$F(t) = 1 - ct^3 + bt^5 + \frac{c^2}{2}t^6 + O(t^7).$$

Thus the fourth-order coefficient vanishes, the fifth-order coefficient is the first potentially new one, and the sixth-order coefficient is already forced by c.

Why the coercive case is different. When $m = 0$, one has $c = 2\lambda_{\min}(H) > 0$. Lemma 1 still gives

$$\log F(t) = -ct + bt^3 + O(t^5),$$

but now exponentiation immediately produces an even term:

$$F(t) = 1 - ct + \frac{c^2}{2}t^2 + \left(b - \frac{c^3}{6}\right)t^3 + O(t^4). \tag{6}$$

Thus the assertion "the remainder is one order higher" is genuinely hypocoercive: it relies not only on the oddness of $\log F$, but also on the fact that its leading order is at least three. In the coercive case the quadratic term $c^2t^2/2 = 2\lambda_{\min}(H)^2t^2$ is unavoidable.

The leading law (1) follows by squaring the finite-dimensional short-time result of F. Achleitner, A. Arnold, and E. A. Carlen, *J. Differential Equations* **371** (2023), 83–115, Theorem 2.7.